Theoretical Physics 2 - Blatt 06

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15. Orbital Angular Momentum Operatos and Spherical Harmonics We consider the orbital angular momentum operators $\hat{L}_j = \sum_{k,l=0}^{3} \epsilon_{jkl}\hat{x}_k\hat{p}_l$.

1. Write the operators $\hat{L}_j$, $\sum_j \hat{L}_j^2$, $\hat{L}_\pm = \hat{L}_1 \pm i\hat{L}_2$ in the **in the position representation** (in cartesian coordinates).

Calculation

We are given $\hat{L}_j = \sum_{k,l=1}^{3} \epsilon_{j,k,l}\hat{x}_k\hat{p}_l$
In cartesian coordinates $\hat{p}_l = -i\hbar\frac{\partial}{\partial x_l}$
so we get $\hat{L}_j = -\hbar\sum_{k,l=1}^{3} \epsilon_{jkl}\frac{\partial}{\partial x_L}$
Single component:
$j = 1$

Calculation

$\hat{L}_1 = \hat{x}_2\hat{p}_3 - \hat{x}_3\hat{p}_2 = -i\hbar(x_2\frac{\partial}{\partial x_3} - x_3\frac{\partial}{\partial_2})$

$j = 2$

Calculation

$\hat{L}_2 = \hat{x}_3\hat{p}_1 - \hat{x}_1\hat{p}_3 = -i\hbar(x_3\frac{\partial}{\partial x_1} - x_1\frac{\partial}{\partial_3})$

$j = 3$

Calculation

$\hat{L}_3 = \hat{x}_1\hat{p}_2 - \hat{x}_2\hat{p}_1 = -i\hbar(x_1\frac{\partial}{\partial x_2} - x_2\frac{\partial}{\partial_1})$

Total operator $\hat{L}^2$

Calculation

$\hat{L}^2 = \hat{L}_1^2 + \hat{L}_2^2 + \hat{L}_3^2 = -\hbar^2[(x_2\frac{\partial}{\partial x_3} - x_3\frac{\partial}{\partial x_2})^2 + (x_3\frac{\partial}{\partial x_1} - x_1\frac{\partial}{\partial x_3})^2 + (x_1\frac{\partial}{\partial x_2} - x_2\frac{\partial}{\partial x_1})^2]$

Ladder operator

Calculation

$\hat{L}_\pm = \hat{L}_2 \pm i\hat{L}_2 = -i\hbar(x_2\frac{\partial}{\partial x_3} - x_3\frac{\partial}{\partial x_2}) \pm \hbar(x_3\frac{\partial}{\partial x_1} - x_1\frac{\partial}{\partial x_3})$

2. Express the same operators in **spherical coordinates**, using $x_1 = r\sin\theta\cos\phi$, $x_2 = r\sin\theta\sin\phi$, $x_3 = r\cos\theta$. (e.g., by applying the chain rule).

Calculation

Chainrule $\frac{\partial}{\partial x_i} = \frac{\partial r}{\partial x_i}\frac{\partial}{\partial r} + \frac{\partial \theta}{\partial x_i}\frac{\partial}{\partial \theta} + \frac{\partial \phi}{\partial x_i}\frac{\partial}{\partial \phi}$
Identities: $4 = \sqrt{x_1^2 + x_2^2 + x_3^2}$, $x_1^2 x_2^2 = r^2\sin^2\theta$, $\phi = \arctan\frac{x_2}{x_1}$
derivations

Calculation

derivation of r

Calculation

$\frac{r}{x_1} = \frac{x_1}{3} = \sin\theta\cos\phi$

derivation of ϕ

Calculation

$\frac{\partial \phi}{\partial x_1} = -\frac{x_2}{x_1^2+x_2^2} = -\frac{r\sin\theta\cos\phi}{r^2\sin^2\theta} = -\frac{\sin\phi}{r\sin\theta}$

derivation of θ

Calculation

$\cos\theta = \frac{x_3}{r}$
$\frac{\partial}{\partial x_1}\cos\theta = -\sin\theta\frac{\partial}{\partial x_1} = -\frac{x_3}{r^2}\frac{\partial r}{\partial x_1} = -\frac{x_3}{r^2}\frac{x_1}{r} = -\frac{r\cos\theta r\sin\theta\cos\phi}{r^3}$
$\frac{\partial}{\partial x_1} = \frac{\cos\theta\cos\phi}{r}$

partial derivations by x_i

Calculation

x_1

Calculation

$$\frac{\partial}{\partial x_1} = \sin\theta\cos\phi\phi\frac{\partial}{\partial r} + \frac{\cos\theta\cos\phi}{r}\frac{\partial}{\partial\theta} + \frac{\cos\phi}{r\sin\theta}\frac{\partial}{\partial\phi}$$

x_2

Calculation

$$\frac{\partial}{\partial x_2} = \sin\theta\sin\phi\frac{\partial}{\partial r} + \frac{\cos\theta\cos\phi}{r}\frac{\partial}{\partial\theta} - \frac{\cos\phi}{r\sin\theta}\frac{\partial}{\partial\phi}$$

x_3

Calculation

$$\frac{\partial}{\partial x_3} = \cos\theta\frac{\partial}{\partial r} - \frac{\sin\theta}{r}\frac{\partial}{\partial\theta}$$

Components

Calculation

$$\hat{L}_1 = i\hbar(\sin\phi\frac{\partial}{\partial\theta} + \cot\theta\cos\phi\frac{\partial}{\partial\phi})$$
$$\hat{L}_2 = i\hbar(-\cos\phi\frac{\partial}{\partial\theta} + \cot\theta\sin\phi\frac{\partial}{\partial\phi})$$
$$\hat{L}_3 = -i\hbar\frac{\partial}{\partial\phi}$$

total operator $\hat{L}^2$

Calculation

$$\hat{L}^2 = -\hbar[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial}{\partial\theta}) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}]$$

Calculation

Ladder operator

Calculation

$$\hat{L}_\pm = \hbar e^{\pm i\phi}(\pm\frac{\partial}{\partial\theta} + i\cot\theta\frac{\partial}{\partial\phi})$$

Calculation

3. Formulate the two differential equations that a common eigenstate $\psi_{k,l,m}(r,\theta,\phi)$ of

- $\hat{L}^2$ (with eigenvalue $\hbar^2 l(l+1)$)

Calculation

$\hat{L}^2\psi_{k,l,m} = -\hbar^2[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial\psi_{k,l,m}}{\partial\theta}) + \frac{1}{\sin^2\theta}\frac{\partial^2\psi_{k,l,m}}{\partial\phi^2}] = \hbar^2 l(l+1)\psi_{k,l,m}$
$-[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial\psi_{k,l,m}}{\partial\theta}) + \frac{1}{\sin^2\theta}\frac{\partial^2\psi_{k,l,m}}{\partial\phi^2}] = l(l+1)\psi_{k,l,m}$

- $\hat{L}_3$ (with eigenvalue $\hbar m$)

Calculation

$\hat{L}_3\psi_{k,l,m} = -i\hbar\frac{\partial\psi_{k,l,m}}{\partial\phi} = \hbar m\psi_{k,l,m}$
$\frac{\partial\psi_{k,l,m}}{\partial\phi} = im\psi_{k,l,m}$

must satisfy, **without solving them**

Is it always possible to find a solution that satisfies both differential eqations simultaneously?

Calculation

Yes, because commutator $[\hat{L}^2, \hat{L}_3] = 0$ and therefor have common eigenfunctions.

Calculation

$[\hat{L}^2, \hat{L}_3]$
$= [\hat{L}_1^2, \hat{L}_3] + [\hat{L}_2^2, \hat{L}_3] + [\hat{L}_3^2, \hat{L}_3]$
$= \hat{L}_1[\hat{L}_1, \hat{L}_3] + [\hat{L}_1, \hat{L}_3]\hat{L}_1 + \hat{L}_2[\hat{L}_2, \hat{L}_3] + [\hat{L}_2, \hat{L}_3]\hat{L}_2$
$= \hat{L}_1(-i\hbar\hat{L}_2) + (-i\hbar\hat{L}_2)\hat{L}_1 + \hat{L}_2(i\hbar\hat{L}_1) + (i\hbar\hat{L}_1)\hat{L}_2$
$= 0$

4. The eigenstate $\psi_{k,l,l}(r,\theta,\psi)$ satisfies $\hat{L}_+\psi_{k,l,l} = 0$.
   - Consider the resulting differential equation for the special case $l = 1$, and sovle it.

Calculation

We use $\hat{L}_+$ and $\hat{L}_-$ from (2)
$({}_+Y_l^m = \hbar\sqrt{l(l+1) - m(m+1)}Y_l^m + 1)$
For $l = 1$ holds true: $\hat{L}_+Y_1^1 = 0$ and Y_1^1 is eigenfunction from ${}_3$ with $m = 1$:
$Y_1^1(\theta,\phi) = f(\theta)\exp(i\phi)$
so we plug it in:
$0 = \hat{L}_+Y_1^1$
$= \hbar\exp(i\phi)(\frac{\partial}{\partial\theta} + i\cot\theta\frac{\partial}{\partial\phi})f(\theta)\exp(i\phi)$
$= \hbar\exp(i\phi)(f'(\theta)\exp(i\phi) + i\cot\theta i f(\theta)\exp i\phi)$
So we get: $f'(\theta) - \cot\theta f(\theta) = 0$
$\frac{f'(\theta)}{f(\theta)} = \cot\theta$
Integrate: $\ln f(\theta) = \ln(\sin\theta) + C$
$f(\theta) = C\sin\theta$

Therefore $Y_1^1(\theta,\phi) = C\sin\theta\exp(i\phi)$

- The, using the operator $\hat{L}_-$, derive an explicit form of the spherical harmonics $Y_l^m(\theta,\phi)$, which are defined via $\psi_{k,l,m}(r,\theta,\phi) = R_k(r)Y_l^m(\theta,\phi)$

  Here $R_k(r)$ remeins unspecified but is normalized via $\int\limits_0^\infty r^2dr\,\left|R_(r)\right|^2 = 1$.

Calculation

$\hat{L}_-Y_l^m = \hbar\sqrt{l(l+1)-m(m-1)}Y_l^{m-1}$
for $l=1, m=1$

Calculation

$\hbar\sqrt{2}Y_1^0 = \hat{L}_-Y_1^1$
$= \hbar\exp(-i\phi)(-\frac{\partial}{\partial\theta} + i\cot\theta\frac{\partial}{\partial\phi})C\sin\theta\exp(i\phi)$
$= \hbar\exp(-i\phi)(-C\cos\theta\exp(i\phi) - C\cos\theta\exp(i\phi))$
$= -2\hbar C\cos\theta$
Therefore $y_1^0 = -\sqrt{2}C\cos\theta$ with $C = -\sqrt{\frac{3}{8\pi}}$

for $l=1, m=-1$

Calculation

$\hbar\sqrt{2}Y_1^{-2} = \hat{L}_-Y_1^0$
$= \hbar\exp(-i\phi)(-\frac{\partial}{\partial\theta} + i\cot\theta\frac{\partial}{\partial\phi})\sqrt{\frac{3}{4\pi}}\cos\theta$
$= \hbar\exp(-i\phi)(-\sqrt{\frac{3}{4\pi}}\sin\theta)$
Therefore $Y_1^{-1} = \sqrt{\frac{3}{8\pi}}\sin\theta\exp -i\phi$

5. Insert the spherical harmonics into the differential equations from part (3), and show that they are indeed the corresponding eigenfunctions.

Calculation

Previously $\hat{L}^2Y_1^- = \hbar^2 1(1+1)Y_1^m = 2\hbar^2Y_1^m$ and $\hat{L}_3Y_1^m = \hbar mY_1^m$
Check with $\hat{L}_3$

Calculation

For $\hat{L}_3 = -i\hbar\frac{\partial}{\partial\phi}$ for $Y_1^1 = -\sqrt{\frac{3}{8\pi}}\sin\theta\exp i\phi$
We get $\frac{\partial}{\partial\phi}Y_1^1 = iY_1^1$
therefor $\hat{L}_3Y_1^1 = -i\hbar iY_1^1 = \hbar Y_1^1$ So $m=1$

Calculation

For $Y_1^0 = \sqrt{\frac{3}{4\pi}}\cos\theta$
We get $\frac{\partial}{\partial\phi}Y_1^0 = 0$
therefor $\hat{L}_3 Y_1^0 = 0$ So $m = 0$

Calculation

$Y_1^{-1} = \sqrt{\frac{3}{8\pi}}\sin\theta\exp -i\phi$
We get $\frac{\partial}{\partial\phi}Y_1^{-1} = -iY_1^{-1}$
therefor $\hat{L}_3 Y_1^{-1} = -i\hbar(-i)Y_1^{-1} = -\hbar Y_1^{-1}$ So $m = -1$

Check with $\hat{L}^2$

Calculation

From (2) $\hat{L}^2 = -\hbar^2[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial}{\partial\theta}) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}]$
for $Y_1^0 = A\cos\theta$, $A = \sqrt{\frac{3}{4\pi}}$
$\frac{\partial^2}{\partial\phi^2}Y_1^0 = 0$
$\frac{\partial}{\partial\theta}Y_1^0 = -A\sin\theta$
$\sin\theta\frac{\partial}{\partial\theta}Y_1^0 = -A\sin^2\theta$
$\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial}{\partial\theta}Y_1^0) = -2A\sin\theta\cos\theta$
$\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial}{\partial\theta}Y_1^0) = -2A\cos\theta = -2Y_1^0$
So $\hat{L}^2 Y_1^0 = -\hbar^2(-2Y_1^0) = 2\hbar^2 Y_1^0$
For $Y_1^{\pm1}$ we get the same expression $\hat{L}^2 Y_1^{\pm1} = s\hbar^2 Y_1^{\pm1}$
Therefore $\hat{L}Y_1^m = 2\hbar^2 Y_1^m$ and $l(l+1) = 2$ so $l = 1$

16. Scattering from a Box Ootential, Bound States, and Scattering Resonances We consider a three-dimensional Potential $V(r)$ (in spherical coordinates) $V(r) = \begin{cases} -V_0 & r \leq a \\ 0 & r > a \end{cases}$ with a positive constant V_0.

1. Formulate the **radial Schrödinger equation** for the two regions $r < a$ and $r > a$ in the case $l = 0$ (see Exercise 15), and find the **general solution for** $E > 0$ using the condition $P_l(r \to 0) = 0$, where P_l is the radial part of the wavefunction, i.e. $\psi_{E,l=0,m=0}(\vec{x}) = \frac{P_{l=0}(r)}{r} >_0^0 (\theta,\phi)$. Furthermore, using the condition $\frac{1}{P_0(r)}\frac{dP_0(r)}{dr}\Big|_{r=a-0} = \frac{1}{P_0(r)}\frac{dP_0(r)}{dr}\Big|_{r=a+0}$, derive an equation for $\tan\delta_0$, where δ_0 is the $scatteringphaseshiftfor$ $l = 0$.

Calculation

For $\ell = 0$ we use

$$\psi_{E,0,0}(\mathbf{x}) = \frac{P_0(r)}{r}Y_0^0(\theta,\phi)$$

The radial Schrödinger equation becomes

Calculation

$-\frac{\hbar^2}{2m}\frac{d^2P_0}{dr^2} + V(r)P_0 = EP_0$

Region I: $r < a$

Calculation

$V(r) = -V_0$
$-\frac{\hbar^2}{2m}P_0'' - V_0P_0 = EP_0$
$P_0'' + \frac{2m(E+V_0)}{\hbar^2}P_0 = 0$
Define $K^2 = \frac{2m(E+V_0)}{\hbar^2}$
$P_0'' + K^2P_0 = 0$
General solution:
$P_0^<(r) = A\sin(Kr) + B\cos(Kr)$
Using $P_0(r \to 0) = 0$:
$B = 0$
$\Rightarrow P_0^<(r) = A\sin(Kr)$

Region II: $r > a$

Calculation

$V(r) = 0$
$P_0'' + \frac{2mE}{\hbar^2}P_0 = 0$
Define $k^2 = \frac{2mE}{\hbar^2}$
$P_0'' + k^2P_0 = 0$
Scattering solution:
$P_0^>(r) = B\sin(kr + \delta_0)$

Matching condition at $r = a$

Calculation

$\left.\frac{P_0'}{P_0}\right|_{a-0} = \left.\frac{P_0'}{P_0}\right|_{a+0}$

Inside:

Calculation

$P_0^< = A\sin(Kr)$
$P_0' = AK\cos(Kr)$
$\frac{P_0'}{P_0} = K\cot(Kr)$
$\Rightarrow \left.\frac{P_0'}{P_0}\right|_{a-0} = K\cot(Ka)$

Outside:

Calculation

$P_0^{>} = B\sin(kr+\delta_0)$
$P_0' = Bk\cos(kr+\delta_0)$
$\frac{P_0'}{P_0} = k\cot(kr+\delta_0)$
$\Rightarrow \left.\frac{P_0'}{P_0}\right|_{a+0} = k\cot(ka+\delta_0)$

Matching gives:

Calculation

$K\cot(Ka) = k\cot(ka+\delta_0)$

Solve for $\tan\delta_0$:

Calculation

$\tan(ka+\delta_0) = \frac{k}{K}\tan(Ka)$
Using $\tan(ka+\delta_0) = \frac{\tan(ka)+\tan\delta_0}{1-\tan(ka)\tan\delta_0}$
we obtain
$\tan\delta_0 = \frac{\frac{k}{K}\tan(Ka)-\tan(ka)}{1+\frac{k}{K}\tan(Ka)\tan(ka)}$

with

Calculation

$k = \sqrt{\frac{2mE}{\hbar^2}}, \quad K = \sqrt{\frac{2m(E+V_0)}{\hbar^2}}$

2. **Scattering at high and low energies**

a) What is δ_0 in the limit $E \to \infty$? What is the physical meaning?

Calculation

From part (a) we have

$$\tan\delta_0 = \frac{\frac{k}{K}\tan(Ka) - \tan(ka)}{1 + \frac{k}{K}\tan(Ka)\tan(ka)}$$

with

$$k = \sqrt{\frac{2mE}{\hbar^2}}, \qquad K = \sqrt{\frac{2m(E+V_0)}{\hbar^2}}.$$

In the limit $E \to \infty$, the potential depth V_0 becomes negligible compared to E:

$$E + V_0 \approx E.$$

Therefore

$$K \approx k.$$

Hence

$$\frac{k}{K} \to 1, \qquad Ka \to ka.$$

Insert this into the expression for $\tan\delta_0$:

$$\tan\delta_0 = \frac{\tan(ka) - \tan(ka)}{1 + \tan^2(ka)} = 0.$$

Thus

$$\boxed{\delta_0 \to 0}$$

for $E \to \infty$.

Physical interpretation:

Calculation

At very high energy, the particle is barely affected by the finite potential well. The wavelength is very small compared to the length scale of the potential, and the scattering phase shift vanishes.
Therefore, the potential becomes effectively transparent:

$$\boxed{\delta_0 \to 0}$$

means almost no scattering.

b) Determine δ_0 in the limit $E \to 0$, and calculate the **energy-dependent total cross section** $\sigma(E)$

Find the specific values of V_0 and a that lead to **scattering resonances**.

Show that for small energies, the scattering process reduces to **s-wave scattering** (i.e., $l = 0$).

Calculation

From part (1):

$$\tan\delta_0 = \frac{\frac{k}{K}\tan(Ka) - \tan(ka)}{1 + \frac{k}{K}\tan(Ka)\tan(ka)}$$

with

$$k = \sqrt{\frac{2mE}{\hbar^2}}, \qquad K = \sqrt{\frac{2m(E+V_0)}{\hbar^2}}.$$

For $E \to 0$ we have

$$k \to 0, \qquad K \to \kappa, \qquad \kappa := \sqrt{\frac{2mV_0}{\hbar^2}}.$$

Also

$$\tan(ka) \approx ka.$$

Therefore

$$\tan\delta_0 \approx \frac{\frac{k}{\kappa}\tan(\kappa a) - ka}{1}$$

so

$$\tan\delta_0 \approx k\left(\frac{\tan(\kappa a)}{\kappa} - a\right).$$

Define the scattering length

$$a_s := a - \frac{\tan(\kappa a)}{\kappa}.$$

Then

$$\boxed{\tan\delta_0 \approx -ka_s}$$

and for small k:

$$\boxed{\delta_0 \approx -ka_s}.$$

Total cross section:

Calculation

For s-wave scattering,

$$\sigma(E) = \frac{4\pi}{k^2}\sin^2\delta_0.$$

Using

$$\tan\delta_0 \approx -ka_s$$

and

$$\sin^2\delta_0 = \frac{\tan^2\delta_0}{1+\tan^2\delta_0},$$

we get

$$\sigma(E) = \frac{4\pi}{k^2}\frac{k^2a_s^2}{1+k^2a_s^2}.$$

Therefore

$$\boxed{\sigma(E) = \frac{4\pi a_s^2}{1+k^2a_s^2}}$$

with

$$a_s = a - \frac{\tan(\kappa a)}{\kappa}, \qquad \kappa = \sqrt{\frac{2mV_0}{\hbar^2}}.$$

Scattering resonances:

Calculation

A scattering resonance occurs when the scattering length diverges:

$$a_s \to \infty.$$

This happens when

$$\tan(\kappa a) \to \infty.$$

Therefore

$$\kappa a = \left(n + \frac{1}{2}\right)\pi, \qquad n = 0,1,2,\ldots$$

Since

$$\kappa^2 = \frac{2mV_0}{\hbar^2},$$

this gives

$$\boxed{V_0 a^2 = \frac{\hbar^2}{2m}\left(n + \frac{1}{2}\right)^2 \pi^2}$$

At resonance,

$$\sigma(E) \to \frac{4\pi}{k^2}.$$

Low-energy scattering is s-wave scattering:

Calculation

For small energies, higher partial waves are suppressed.
In general,

$$\delta_l \sim k^{2l+1}.$$

Therefore, for $l > 0$,

$$\delta_l \to 0 \quad \text{faster than} \quad \delta_0.$$

Hence only the $l = 0$ contribution remains relevant:

$$\boxed{\text{low-energy scattering reduces to s-wave scattering.}}$$

c) Show that for certain values of $F_0 a^2$, the box potential becomes **transparent**, i.e. $\sigma = 0$ (**Ramsauer-Townsed effect**).

Calculation

From part (b)(ii), for low energies we found

$$\sigma(E) = \frac{4\pi a_s^2}{1 + k^2 a_s^2}$$

with

$$a_s = a - \frac{\tan(\kappa a)}{\kappa}, \qquad \kappa = \sqrt{\frac{2mV_0}{\hbar^2}}.$$

The potential is transparent if

$$\sigma(E) = 0.$$

This happens if

$$a_s = 0.$$

Therefore

$$a - \frac{\tan(\kappa a)}{\kappa} = 0.$$

Hence

$$\tan(\kappa a) = \kappa a.$$

Thus the transparency condition is

$$\boxed{\tan(\kappa a) = \kappa a}$$

with

$$\kappa a = a\sqrt{\frac{2mV_0}{\hbar^2}}.$$

Therefore the corresponding values of $V_0 a^2$ are determined by

$$\boxed{V_0 a^2 = \frac{\hbar^2}{2m} x_n^2}$$

where x_n are the non-zero solutions of

$$\tan x_n = x_n.$$

Physical interpretation:

Calculation

For these special values of $V_0 a^2$, the scattered wave destructively interferes. Therefore the phase shift effectively gives no scattering contribution, and the total cross section vanishes:

$$\boxed{\sigma = 0}$$

This is the Ramsauer–Townsend effect.

3. **Bound states for s-wave scattering**

   a) Show that the general solution of the radial Schrödinger equation for $E < 0$ ha the form $T_{t=0}(r) = \begin{cases} c_1 \sin(Kr) & r \leq a \\ c_2 \exp(-\rho r) & r > a \end{cases}$ and determine ρ and K.

   b) Using the continuity conditions at $r = a$, show that ρ satisfies $\tan(Ka) = \frac{K}{\rho}$. Discuss this equation:

      - Determine the number of bound s-states as a function of the potential depht (for fixed a)

- Show in particula that no bound states exist if the potential is too shallow
- Finally, compare the bound states for $E \to 0^-$ with the scattering resonances in the case $E \to 0^+$ from part (2)(b)